

Count-System Interoperability: Vendor Exports To RCOUNT

Gio Della-Libera

Draft — June 28, 2026

Abstract

Election results are published through many local workflows: election-night exports, statements of votes, canvass reports, vendor reports, spreadsheets, PDFs, and common-data-format files. RCOUNT should not pretend those sources are already neutral ledgers. It should make adapter work explicit. This paper defines the first V.9 interoperability slices: a statement-of-votes CSV adapter and a small NIST Election Results Reporting CDF-style JSON adapter. Both normalize contest, reporting-unit, selection, residual, status, and jurisdiction-total evidence into ordinary RCOUNT packages while preserving the source file hash. The adapters are intentionally small, but they establish the contract future jurisdiction-specific adapters should follow: parse, normalize, hash the original source, verify the resulting package, and label any claim the adapter cannot make.

1 The Interoperability Problem

RCOUNT is only useful if it can meet election offices where their public records already are. Some offices publish machine-readable result exports. Others publish statements of votes as spreadsheets, PDFs, precinct reports, county canvass files, or vendor-specific artifacts. The EAC describes canvass and certification as the process by which unofficial results become official public results [U.S. Election Assistance Commission, 2026]. RCOUNT’s role is narrower: it normalizes declared public evidence into a package that can be mechanically rechecked.

The adapter layer is therefore not a cosmetic import step. It is where source meaning becomes explicit. A field named “total” might mean votes for one selection, ballots cast in a contest, ballots accepted in a batch, or a jurisdiction-wide summary. A status label might mean election-night unofficial, canvassed, recounted, amended, or certified. An adapter must not guess silently. It must map each source field to a typed RCOUNT object or leave the source claim outside the verifier.

2 Adapter Contract

An RCOUNT import adapter should satisfy five rules.

1. Preserve the original source bytes and register them in the source index.
2. Normalize only fields whose meaning the adapter understands.
3. Emit ordinary RCOUNT records rather than an adapter-only side channel.
4. Run the normal verifier after import, so arithmetic and source hashes share the same transcript surface.
5. Name the unsupported claims explicitly: eligibility, voting-system certification, ballot-marking correctness, and legal certification remain outside the adapter.

This mirrors the direction of NIST election common-data-format work, which separates interoperability schemas from the legal judgment that a result is official [National Institute of Standards and Technology, 2019, 2026]. RCOUNT can ingest such schemas later, but the package invariant is the same: source evidence is hashed, normalized evidence is typed, and verifier equations describe exactly what was checked.

3 Implemented Import Slices

The first implemented V.9 adapters are intentionally small. The statement CSV command is:

```
rcount import-statement-csv <csv> <output-dir>
```

The CSV uses one row per contest, reporting unit, selection, and status. It has four column families:

- Contest columns: id, title, and vote-for cardinality.
- Selection columns: id, label, and candidate or write-in-bucket kind.
- Unit columns: reporting-unit id, reporting-unit kind, and parent jurisdiction.
- Count columns: status, selection votes, undervotes, overvotes, blanks, and counted ballots.

The adapter stores the original statement file as source evidence and gives it a stable statement-CSV source label. It also produces normal contest, reporting-unit, and summary NDJSON files. A package created this way is verified by `rcount verify`; the source-hash check is reported as the same source hash equation used by synthetic fixtures.

The second command imports a small NIST ERR/CDF-style JSON fixture:

```
rcount import-nist-cdf-json <json> <output-dir>
```

That adapter reads CDF-shaped group units, contests, contest selections, vote counts, other-count residuals, and result status fields. It maps recognized group-unit types to RCOUNT reporting-unit kinds, preserves the source file under a stable CDF source label, and computes residual-aware counted ballots from selection votes, undervotes, overvotes, and blank contests.

4 Failure Modes

The adapter rejects missing required cells, bad status labels, unsupported reporting-unit kinds, invalid numeric fields, duplicate selection rows inside a summary, and conflicting repeated metadata. After import, the ordinary verifier catches the deeper count errors: selection sums that do not equal counted ballots, jurisdiction totals that do not equal precinct totals, unknown selections, negative counts, and source-file tampering.

This two-layer structure matters. Parser failures are evidence-meaning failures: the source cannot be safely normalized. Verifier failures are package-consistency failures: the source was normalized, but its declared equations do not hold. Keeping those categories separate makes later vendor adapters easier to audit.

5 Boundaries

The statement CSV adapter does not certify an election. It does not prove that a voting system recorded voter intent correctly, that voters were eligible, that every ballot was lawfully accepted, or that a jurisdiction’s official board has completed its statutory duties. It only proves that a source file was preserved, that understood fields were normalized into RCOUNT records, and that the resulting package passes the mechanical RCOUNT equations.

The CDF-style adapter also does not claim to be a complete NIST Election Results Reporting implementation. Its purpose is to establish the RCOUNT import pattern before larger jurisdiction-specific adapters arrive.

6 Conclusion

V.9 turns interoperability from an aspiration into a repeatable pattern. A public result export can enter RCOUNT through a named adapter, keep its original source hash, and emerge as the same package shape used by canvass arithmetic, batch accounting, CVR reconciliation, RLA replay, and certification evidence matrix work.

The next depth is jurisdictional breadth: real statement formats and negative cases for ambiguous fields. Each new adapter should make one promise only after it can show the source bytes, the normalized records, and the verifier transcript that connect them.

References

Gio Della-Libera. Rcount substrate specification. Technical report, Apportionment Research, 2026.

National Institute of Standards and Technology. Election results common data format specification: Revision 2.0. <https://www.nist.gov/publications/election-results-common-data-format-specificationrevision-20>, 2019. Accessed 2026-05-13.

National Institute of Standards and Technology. Implementation guidance for common data formats. <https://nvlpubs.nist.gov/nistpubs/gcr/2024/24-058/NIST.GCR.24-058.html>, 2026. Accessed 2026-05-13.

U.S. Election Assistance Commission. Election results, canvass, and certification. <https://www.eac.gov/election-officials/election-results-canvass-and-certification>, 2026. Accessed 2026-05-13.